

SOP-002: Synthesis of Highly-Branched SAF

Target Molecule: CCCC(CC)(CC)CCC(C)C (Heavily Branched C15 Alkene)

Origin: Shodh AI Engine 4 (Autonomous Retrosynthesis)

Safety & PPE

Hazards: Grignard reagents, including ethylmagnesium bromide, react violently and explosively with water and air.

Requirements: Step 1 must be performed under an inert atmosphere (nitrogen/argon) using dry solvents. Exothermic reactions expected; have an ice bath ready.

Reagents & Volumes

Reagent / Material	Volume / Amount	Notes
Standard dry solvent	300.0 uL total	e.g., THF/ether; 100.0 uL per reaction vial
Heptan-3-one	42.9 uL	Ketone
Ethylmagnesium bromide	51.4 uL	Grignard reagent
Acetone	25.7 uL	Branched proxy
Sulfuric acid (H2SO4)	20.0 uL	Dehydration reagent
Hydrogenation reagent	90.0 uL	e.g., H2 gas proxy / donor

Step-by-Step Bench Procedure

Step 1: Grignard Alkylation (Forming the Branches)

- Under inert gas, prepare Reaction Vial 1. Add 100.0 uL of dry solvent.
- Add 42.9 uL of heptan-3-one.
- Carefully, drop-wise, add 51.4 uL of ethylmagnesium bromide. Monitor for exothermic heat.
- Add 25.7 uL of acetone.
- Agitate 5 times. Stir for 10 minutes to allow the carbon-carbon bonds to form.

Step 2: Acid-Catalyzed Dehydration (Creating the Double Bond)

- Prepare a clean Reaction Vial 2. Add 100.0 uL of solvent.
- Transfer exactly 80.0 uL of the crude alcohol intermediate from Vial 1 into Vial 2.
- Slowly add 20.0 uL of sulfuric acid to strip the oxygen and force the C=C double bond.
- Agitate 5 times.
- Stir for 10 minutes.

Step 3: Catalytic Hydrogenation (Final Stabilization)

- 1 Prepare a clean Reaction Vial 3. Add 100.0 uL of solvent.
- 2 Transfer exactly 80.0 uL of the crude alkene from Vial 2 into Vial 3.
- 3 Introduce the hydrogenation reagent (90.0 uL) over a catalyst to stabilize the final branching structure.
- 4 Agitate 5 times.
- 5 Stir for 10 minutes.

Quality Control

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Perform GC-MS (Gas Chromatography-Mass Spectrometry) on the final product.
Verify: Look for the absence of OH peaks, proving successful dehydration, and verify the highly branched hydrocarbon structure.